

CLINICAL INVESTIGATION OPEN ACCESS

Impact of Comprehensive Medication Reviews on Medicare Beneficiaries With Type 2 Diabetes

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ABSTRACT

Background: Type 2 diabetes (T2DM) management, particularly adherence to oral antidiabetic drugs (OAD), is a key focus of the Part D Medicare Medication Therapy Management (MTM) program. Despite potential benefits, evidence on the effectiveness of MTM, specifically Comprehensive Medication Reviews (CMRs), remains mixed. To evaluate the effectiveness of CMRs on patient outcomes, we conducted the largest CMR evaluation to date among Medicare beneficiaries with T2DM.**Methods:** We conducted a retrospective observational study using Medicare claims and MTM data from 2013 to 2019 to examine the impact of CMR on OAD adherence as measured by the proportion of days covered (PDC), statin utilization, emergency department (ED) visits, and hospitalization. A difference-in-difference (DiD) approach with inverse probability treatment weighting was used to reduce confounding.**Results:** Although CMR recipients and nonrecipients both experienced a reduction in PDC during the follow-up, CMR recipients experienced 2.4% (95% CI, 1.9%–2.9%) less of a decline than nonrecipients, and 2.0% (95% CI, 1.2%–2.8%) fewer CMR recipients were considered non-adherent after CMR delivery. The proportion of CMR recipients using statin increased 1.4% (95% CI, 0.7%–2.2%) relative to nonrecipients after CMR delivery. The proportion of CMR recipients who experienced an ED visit declined by 0.3% (95% CI, –0.6% to 1.2%) while the proportion of hospitalization increased 1.9% (95% CI, 1.1%–2.8%) following CMR delivery, relative to nonrecipients.**Conclusion:** Patients with T2DM receiving a CMR experienced improved OAD adherence and statin utilization compared to nonrecipients. Our results demonstrate the potential benefits of collaborating with pharmacists to improve diabetes care.

1 | Introduction

Type 2 diabetes (T2DM) presents a significant health challenge in the United States, leading to substantial mortality, morbidity, and healthcare costs [1]. Nonadherence to oral antidiabetic drugs (OAD) exacerbates these issues, resulting in poor glycemic control,

increased morbidity, and escalated healthcare expenses [2, 3]. In response to T2DM's rising incidence, the Centers for Medicare & Medicaid Services (CMS) have launched several programs to improve diabetes care and adherence [4]. These efforts are part of a comprehensive strategy targeting the management of nine major chronic conditions [5]. A pivotal aspect of this strategy is

Abbreviations: CMR, Comprehensive Medication Review; DiD, difference-in-differences; MTM, Medication Therapy Management; OAD, oral antidiabetic drugs; PDC, proportion of days covered; T2DM, type 2 diabetes.

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Summary

- Key points:
 - Since the implementation of Medication Therapy Management (MTM) services as a requirement under the Medicare Part D benefit in 2006, evidence regarding their effectiveness in diabetes care has remained inconclusive.
 - Using Medicare data from 2013 to 2019, our study, the largest to date, demonstrates significant improvements in medication use quality among patients who received a Comprehensive Medication Review (CMR), including higher oral antidiabetic drugs (OAD) adherence rates and increased utilization of statin therapy.
 - These outcomes emphasize the pivotal role of CMR interventions and pharmacist involvement in healthcare teams for optimizing diabetes management and improving patient health outcomes.
- Why does this paper matter?
 - This study addresses a critical gap in the evidence by providing the largest and most comprehensive evaluation of the Medicare MTM/CMR program's effectiveness in improving OAD adherence, statin use, and healthcare utilization among older adult patients with diabetes, offering insights that can guide policy and practice improvements.
 - Additionally, the findings highlight the significant role of pharmacists in enhancing diabetes management, while also identifying opportunities to optimize MTM delivery and increase CMR uptake to maximize patient benefits.

the Medicare Part D Medication Therapy Management (MTM) program [6]. The MTM program aims to enhance medication safety and efficacy, thereby aiding beneficiaries in more effectively managing their medications [6, 7]. Central to this initiative are Comprehensive Medication Reviews (CMRs), which provide detailed medication assessments and create customized medication action plans to optimize the benefits of drug therapy [8]. A key element of the CMR process is the assessment and resolution of medication therapy problems (MTPs) that patients may experience [9, 10]. MTPs may include the use of unnecessary medication as well as the need for additional medication (e.g., including statin medications when indicated by guidelines) [10–12]. Other types of MTPs include problems related to effectiveness and safety as well as medication adherence [10, 11]. MTM Part D plan sponsors are required to offer an annual CMR to patients who meet the plan's eligibility requirements, with diabetes included as a priority condition for MTM eligibility [13].

Despite the adoption of MTM into the Part D program in 2006, the cumulative evidence of MTM effectiveness has been previously shown to be “inconclusive” [14]. This uncertainty persists, with more recent studies finding either no benefit of MTM on medication adherence or improvement only for antihypertensive and hyperlipidemia medications but not for asthma, depression, or diabetes medications [15, 16]. To date, however, MTM studies have concentrated on evaluations conducted by individual health systems, with variability in the way in which MTM is delivered and among limited samples [14].

Given unsettled evidence of the benefits of the MTM program on diabetes [14–17], the objective of this study was to examine the effectiveness of receiving a CMR in the Medicare Part D MTM program among a large generalizable population of patients with diabetes. Specifically, we focus first on the impact of a CMR on two medication quality measures from the Medicare Star Ratings program, adherence to OAD, and concurrent use of statins. Given a strong association found between diabetes medication adherence and improved glycemic control as well as reduced utilization of intensive health services, we further examine the effectiveness of receiving a CMR on inpatient and emergency department (ED) encounters [18, 19]. This study presents the largest evaluation of MTM and CMR benefits in diabetes to our knowledge, leveraging a large, comprehensive administrative claims dataset of Medicare beneficiaries employing a strong observational study design.

2 | Methods

2.1 | Study Design

This study utilized a retrospective observational cohort study design, employing a difference-in-differences (DiD) approach. We analyzed Medicare beneficiaries enrolled in the Part D MTM program diagnosed with T2DM using ICD-9 codes 250.x0, 250.x2 and ICD-10 code E11, between 2014 and 2018. To ensure a 1-year baseline and post-CMR follow-up period for all participants, data from 2013 to 2019 were used. The participants were categorized into two groups: an intervention group that received a CMR (CMR recipients) and a control group of eligible individuals who were eligible for, but did not receive a CMR (CMR nonrecipients). The index date for CMR recipients was set as the first date of receiving a CMR and for nonrecipients as the first date of a CMR offer following CMR eligibility. We used a new user study design to isolate the effect of a CMR by excluding patients with a CMR during the prior year of observation. A 1-year pre-period before the index date was established as the baseline period with outcomes measured over a 1-year post-period.

2.2 | Data Source

We leveraged two different Medicare data sources for this study. First, we used a 20% random sample of Medicare beneficiary administrative claims histories for inpatient, outpatient, and prescription drug utilization [20]. Patients in the 20% sample were then merged with MTM delivery files which contain detailed information regarding MTM eligibility, CMR offer, CMR delivery, and dates associated. This data allowed for a strong observational study approach among a large sample of Medicare beneficiaries using well-established epidemiological methods. Medicare claims data used in our study were obtained from the CMS for research purposes.

2.3 | Study Population

To qualify for inclusion in the study, patients had to meet several criteria. First, the study population was restricted to patients aged 66 and older at the time of cohort entry to examine the effect of the program among older adults. An age of 66

or older on the index date was set to ensure all patients were at least 65 years old and qualified for MTM during the 1-year baseline period. The sample was also restricted to OAD users with confirmed diabetes by requiring patients to have at least one diagnostic claim of T2DM and two or more OAD prescription fills during the baseline period. This approach ensures high sensitivity and specificity of diabetes diagnosis [21]. The requirement to include two or more medications for inclusion is also consistent with the Medicare Star Ratings methodology for calculating medication adherence. We did not require OAD prescriptions in the post-period to accommodate potential discontinuation due to poor adherence. Patients who did not maintain continuous enrollment in Medicare Parts A, B, and D for 12 months prior to the cohort entry date were excluded from the study to ensure data availability during the 1-year baseline period before the index date and to reduce the likelihood of loss to follow-up. Patients using injectable antidiabetic medications, including insulin and glucagon-like peptide-1 (GLP-1) receptor agonists, were excluded as prior research has shown inconsistency in reporting day supply for injectable medications, which is an important data element in the calculation of medication adherence [22, 23].

2.4 | Outcomes

In the study, the proportion of days covered (PDC) was used for OAD as a metric to assess medication adherence for each patient. PDC is defined as the proportion of days during a specified period that an individual has access to their medications [24]. We calculated the PDC for 1 year before the cohort entry date and for 1 year afterward, starting from the date of the first OAD prescription in each period. If no prescriptions were observed in the post-period, the PDC was set to 0, reflecting nonadherence throughout the post-period. A PDC threshold of 80% was used to denote adherence. This process of measurement aligns with the measurement of diabetes adherence in the Medicare Star Ratings program [4, 22]. In addition to adherence, the statin utilization rate was also examined before and after the cohort entry date for both groups. The statin utilization rate was defined as the proportion of patients who had at least one statin prescription record during the observation period following the Medicare Star Ratings technical specifications [4, 22]. Finally, to examine the impact of the program at preventing potentially high resource health services, evaluations were conducted of the impact of CMRs on both count and probability of an inpatient admission and an ED visit, respectively, between the 1-year pre- and post-periods of observation.

2.5 | Statistical Analysis

To mitigate the potential for confounding factors influencing our findings, the inverse probability treatment weighting (IPTW) approach was adopted. Demographics were included, such as age, sex, race/ethnicity, geographic region, urban/rural, income status, end-stage renal disease (ESRD), index year, copayment status, as well as health conditions including heart failure, depression, fluid and electrolytes disorder, hypertension, hypothyroidism, obesity, paralysis, pulmonary

circulation disorder, and valve disease as confounders for adjustment. This technique helped to balance covariates between the intervention and control groups [25, 26]. The propensity score (PS) for receiving a CMR service was calculated through multivariable logistic regression. For individuals in the intervention group, weights were assigned as the inverse of the PS ($1/PS$), where PS is the probability of receiving a CMR. Conversely, for the control group, weights were assigned as the inverse of the probability of not receiving a CMR ($1/[1 - PS]$) [25, 26]. This approach effectively creates a pseudo population in which the covariates are balanced, approximating the conditions of a randomized study [26]. To assess the success of this balancing act, we provided standardized mean differences (SMD) for each covariate after applying IPTW. An SMD of less than 10% is viewed as evidence of a minimal difference between the two groups.

For the adherence analysis, the DiD estimates in the PDC were calculated using the model: $DC = \beta_0 + \beta_1 \times \text{Time} + \beta_2 \times \text{CMR} + \beta_3 \times (\text{Time} \times \text{CMR}) + \epsilon$, where β_3 represents the DiD estimate [27]. Generalized estimating equations (GEE) were employed and incorporated with IPTW to generate DiD estimates and to address the issue of repeated measurements for the same individuals across the pre- and post-periods. Dichotomous outcomes such as the proportion of patients being adherent to OAD or initiating statin treatment were modeled using a binomial distribution and log link function specifying an independent correlational structure. The significance level for our study was set at 0.05. Data analysis was conducted using SAS software version 9.4 and RStudio version 3.

2.6 | Subgroup and Sensitivity Analyses

To evaluate the effectiveness of CMR across diverse populations, the study included several subgroup analyses based on age (≤ 75 vs. > 75), sex (male vs. female), race/ethnicity (non-Hispanic White, Black/African American, Asian/Pacific Islander, Hispanic, American Indian/Alaska Native), geographic region (Northeast, South, Midwest, West, other), urban versus rural settings, and enrolling into subsidized Medicare programs as a reflection of income subsidy status (Medicaid enrollment, qualified Medicare beneficiary, specified low-income Medicare beneficiary, qualified disabled working individual, non-Medicaid or unknown). These analyses supported an understanding of the performance of CMR in addressing the specific needs of different patient groups. Additionally, to test the robustness of the study results, a sensitivity analysis was conducted by adjusting the PDC threshold from 80% to 90% and using a 180-day pre-index baseline and a 180-day post-index period for calculating the rate and frequency of hospitalizations and ED visits.

3 | Results

A total of 8,973,005 individuals who were offered a CMR service between 2014 and 2018 were initially enrolled in the sample. After applying the inclusion criteria to the 20% sample cohort, 35,816 CMR recipients were enrolled and 71,114 nonrecipients remained in the study cohort. The process of cohort selection

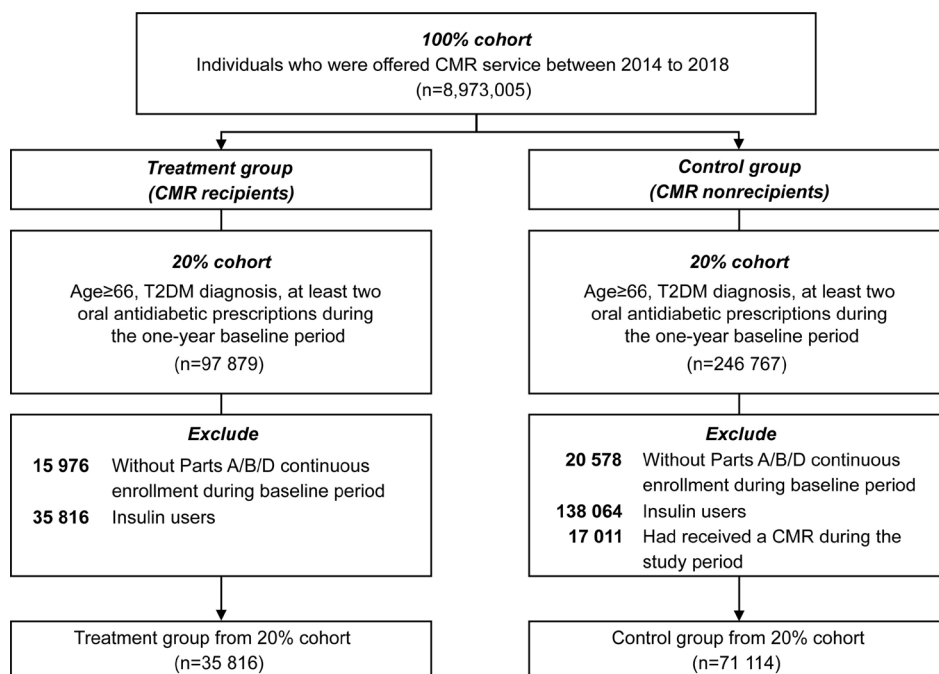


FIGURE 1 | Identification of eligible patients and development of cohorts in the study. A 20% sample of Medicare beneficiaries was linked with Medication Therapy Management (MTM) files. Patients were classified into two groups: those who received a Comprehensive Medication Review (CMR) during the study period (CMR recipients) and those who did not (CMR nonrecipients). Inclusion and exclusion criteria were subsequently applied to define the final cohort for analysis.

is detailed in Figure 1. Within this cohort, the median age was 74 years, with females comprising 56.4% of the population, and 72.3% identifying as White. Table 1 presents the baseline characteristics of the two groups of participants, both before and after the application of IPTW. Before IPTW, CMR recipients were younger, with a lower percentage aged 86 years or over (7.5% vs. 11.6% for nonrecipients), but had higher rates of comorbidities such as congestive heart failure (14.2% vs. 10.7%), hypothyroidism (33.6% vs. 20.6%), paralysis (46.6% vs. 30.3%), and valvular disease (43.1% vs. 23.3%) compared to nonrecipients. After IPTW, the standardized mean differences were all less than 10%, indicating a good balance between the CMR recipient group and the CMR nonrecipient groups.

3.1 | PDC

The PDC during the 12-month baseline period was 85.5% for CMR recipients and 84.7% for nonrecipients. After receiving or being offered a CMR, CMR recipients' PDC declined 11.4% (95% CI, 10.9%–11.8%) to 74.1% versus a 13.8% decline (95% CI, 13.5%–14.0%) to 70.9% for nonrecipients. Despite a decline in adherence in both groups, the DiD estimate suggests that CMR delivery resulted in 2.4% (95% CI, 1.9%–2.9%) higher PDC among CMR recipients comparatively. Details of results for each outcome are presented graphically in Figure 2 and numerically with all DiD results in Data S1.

When employing a cutoff value of 80% for the PDC to define adherence, 73.5% of CMR recipients and 71.5% were adherent to OAD during the baseline period. However, the proportion of patients labeled as adherent declined during the post-period in both groups, with the proportion of patients being labeled

adherent declining by 9.3% (95% CI, 8.7% to 10.0%) among CMR recipients and by 11.3% (95% CI, 10.9% to 11.7%) among nonrecipients. This represents a 2.0% (95% CI, 1.2% to 2.8%) higher proportion of CMR recipients deemed adherent following CMR delivery relative to nonrecipients (Figure 2).

3.2 | Statin Utilization

During the baseline period, the proportion of patients using statin medication was 86.4% for CMR recipients and 86.3% for CMR nonrecipients. After CMR delivery, the proportion of CMR recipients using statin medication increased to 88.1% while the rate increased to 86.5% among nonrecipients. The increase in the proportion of patients using statin medication from baseline to the post-period was 1.7% (95% CI, 1.1%–2.4%) for CMR recipients and 0.3% (95% CI, –0.1% to 0.7%) for nonrecipients. The DiD estimate suggests that CMR implementation improved the proportion of patients with diabetes who used statin medication by 1.4% (95% CI, 0.7%–2.2%). (Figure 2).

3.3 | ED Visit

In regards to ED visits, 35.8% of CMR recipients and 37.0% of nonrecipients had experienced an ER visit during the baseline period. This proportion decreased to 31.8% of CMR recipients and 33.3% of nonrecipients during the post-period, representing a 4.1% (95% CI, 3.3%–4.8%) and 3.7% (95% CI, –3.3% to 4.2%) reduction, respectively. The DiD estimate for the proportion of ED visits was nonsignificant (DiD=0.3% [95% CI, –0.6% to 1.2%]) (Figure 3). Similarly, when considering ED visits as a count variable, our DiD estimate does not suggest a significant difference

TABLE 1 | Baseline characteristics of study participants before and after inverse probability treatment weight.

Variables	Unweighted cohort			Weighted cohort ^a		
	CMR nonrecipients	CMR recipients	SMD	CMR nonrecipients	CMR recipients	SMD ^b
	N = 71,114	N = 35,816		%	%	
Sociodemographic						
Age (%)						
65–70	19,854 (27.9)	10,397 (29.0)	0.155	28.8	30.5	0.039
71–75	19,068 (26.8)	10,861 (30.3)		27.8	27.3	
76–80	14,442 (20.3)	7584 (21.2)		20.5	20.2	
81–85	9509 (13.4)	4278 (11.9)		12.8	12.3	
86+	8241 (11.6)	2696 (7.5)		110.2	9.7	
Sex (%)						
Male	31,248 (43.9)	15,381 (42.9)	0.020	43.6	43.5	0.002
Female	39,866 (56.1)	20,435 (57.1)		56.4	56.5	
Race (%)						
Non-Hispanic White	50,731 (71.3)	26,558 (74.2)	0.093	72.4	72.9	0.013
Black/African American	8045 (11.3)	4246 (11.9)		11.5	11.4	
American Indian/Alaska Native	8361 (11.8)	3426 (9.6)		11.0	10.6	
Others	3977 (5.6)	1586 (4.4)		5.2	5.1	
Region (%)						
Northeast	13,104 (18.4)	6670 (18.6)	0.048	18.5	18.5	0.010
South	28,289 (39.8)	14,204 (39.7)		39.8	40.1	
Midwest	18,455 (26.0)	9807 (27.4)		26.3	25.9	
West	11,074 (15.6)	5045 (14.1)		15.1	15.2	
Other	192 (0.3)	90 (0.3)		0.3	0.3	
Urban (%)						
Yes	56,438 (79.4)	27,530 (76.9)	0.060	78.5	78.4	0.002
No	14,676 (20.6)	8286 (23.1)		21.5	21.6	
Income status (%)						
Medicaid enrollment	23,943 (33.7)	9923 (27.7)	0.140	31.4	30.6	0.018
QMB enrollment	2433 (3.4)	1527 (4.3)		3.7	3.7	
SLMB/QI enrollment	1480 (2.1)	1002 (2.8)		2.3	2.3	
QDWI individual	863 (1.2)	590 (1.6)		1.4	1.4	
Non-Medicaid	42,395 (59.6)	22,773 (63.6)		61.2	62.0	
ESRD (%)						
Yes	1130 (1.6)	443 (1.2)	0.030	1.5	1.4	0.003

(Continues)

TABLE 1 | (Continued)

Variables	Unweighted cohort			Weighted cohort ^a		
	CMR nonrecipients	CMR recipients	SMD	CMR nonrecipients	CMR recipients	SMD ^b
	N = 71,114	N = 35,816		%	%	
No	69,984 (98.4)	35,373 (98.8)		98.5	98.6	
Index year, <i>n</i> (%)						
2014	35,997 (50.6)	6100 (17.0)	0.813	39.4	39.8	0.008
2015	13,239 (18.6)	7467 (20.8)		19.3	19.1	
2016	9155 (12.9)	6971 (19.5)		15.0	14.9	
2017	7793 (11.0)	8640 (24.1)		15.4	15.3	
2018	4930 (6.9)	6638 (18.5)		10.8	10.8	
Copayment status (%)						
Catastrophic coverage	27,398 (38.5)	15,532 (43.4)	0.170	39.9	38.8	0.025
Coverage gap exposure	25,066 (35.2)	9806 (27.4)		32.8	33.9	
No coverage gap exposure	18,650 (26.2)	10,478 (29.3)		27.3	27.3	
Health conditions						
Congestive heart failure, <i>n</i> (%)	7612 (10.7)	5094 (14.2)	0.430	11.8	11.6	0.006
COPD, <i>n</i> (%)	4888 (6.9)	2432 (6.8)	0.003	6.8	6.6	0.006
Depression, <i>n</i> (%)	3556 (5.0)	2889 (8.1)	0.120	6.0	6.0	0.001
Fluid and electrolyte disorder, <i>n</i> (%)	7216 (10.1)	4625 (12.9)	0.087	11.8	10.2	0.052
Hypertension, <i>n</i> (%)	3791 (5.3)	2592 (7.2)	0.079	6.0	6.0	<0.001
Hypothyroidism, <i>n</i> (%)	14,650 (20.6)	12,029 (33.6)	0.295	24.8	25.1	0.008
Obesity, <i>n</i> (%)	5396 (7.6)	5270 (14.7)	0.228	10.0	10.0	<0.001
Paralysis, <i>n</i> (%)	21,522 (30.3)	16,705 (46.6)	0.342	35.9	35.4	0.010
Pulmonary circulation disorder, <i>n</i> (%)	7978 (11.2)	6064 (16.9)	0.165	12.8	13.3	0.016
Valvular disease, <i>n</i> (%)	16,552 (23.3)	15,427 (43.1)	0.430	30.1	29.5	0.014

Abbreviations: CMR, Comprehensive Medication Review; COPD, congestive obstructive pulmonary disease; ESRD, end-stage renal disease; QDWI, qualified disabled and working individuals; QI, qualifying individual; QMB, qualified Medicare beneficiary; SLMB, specified low-income Medicare beneficiary; SMD, standardized mean difference.

^aThe cohort number was weighted by the inverse probability treatment weight approach to balance the covariates, ensuring a reduction in the bias due to confounding variables. Each individual in the study was assigned a weight calculated as the inverse of the probability of receiving the treatment given the observed covariates.

^bA SMD value less than 0.1 indicates a negligible difference in the means, representing a balanced distribution of the characteristic between the groups under comparison.

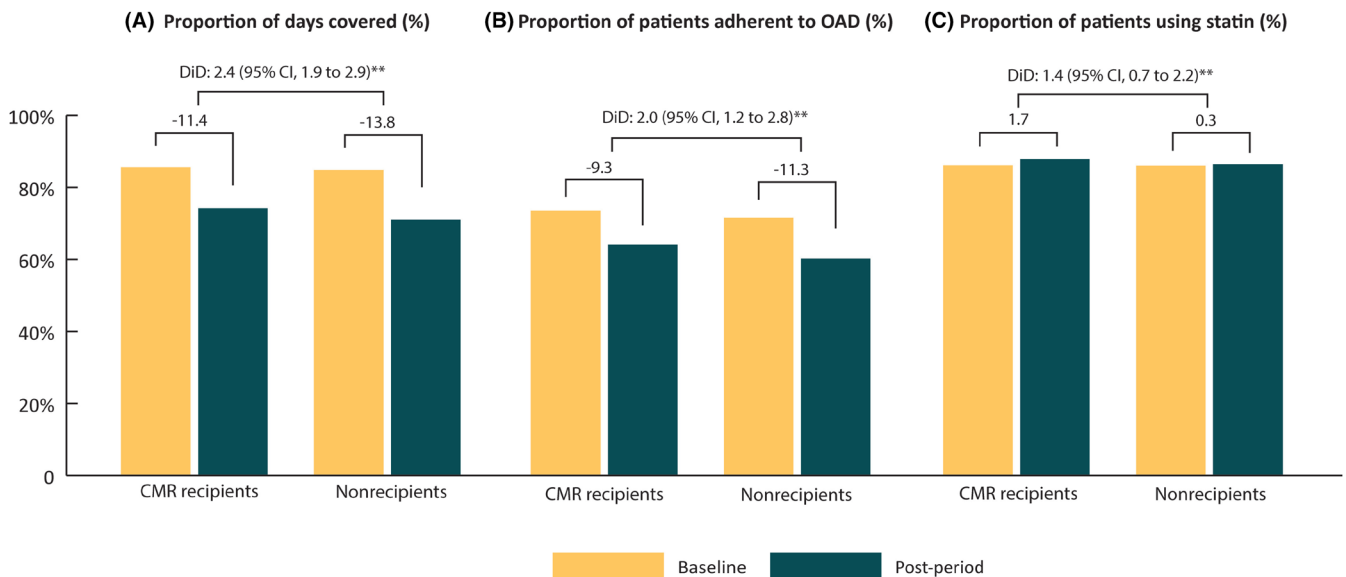


FIGURE 2 | Impact of Comprehensive Medication Review (CMR) on oral antidiabetic drugs (OAD) adherence and statin use. The figure presents the impact of CMR on three outcomes: (A) proportion of days covered (PDC) by OAD, (B) proportion of patients adherent to OAD, and (C) proportion of patients using statins. The difference-in-differences (DiD) estimates and confidence intervals (CIs) indicate the changes between CMR recipients and nonrecipients over time. ** p -value < 0.001.

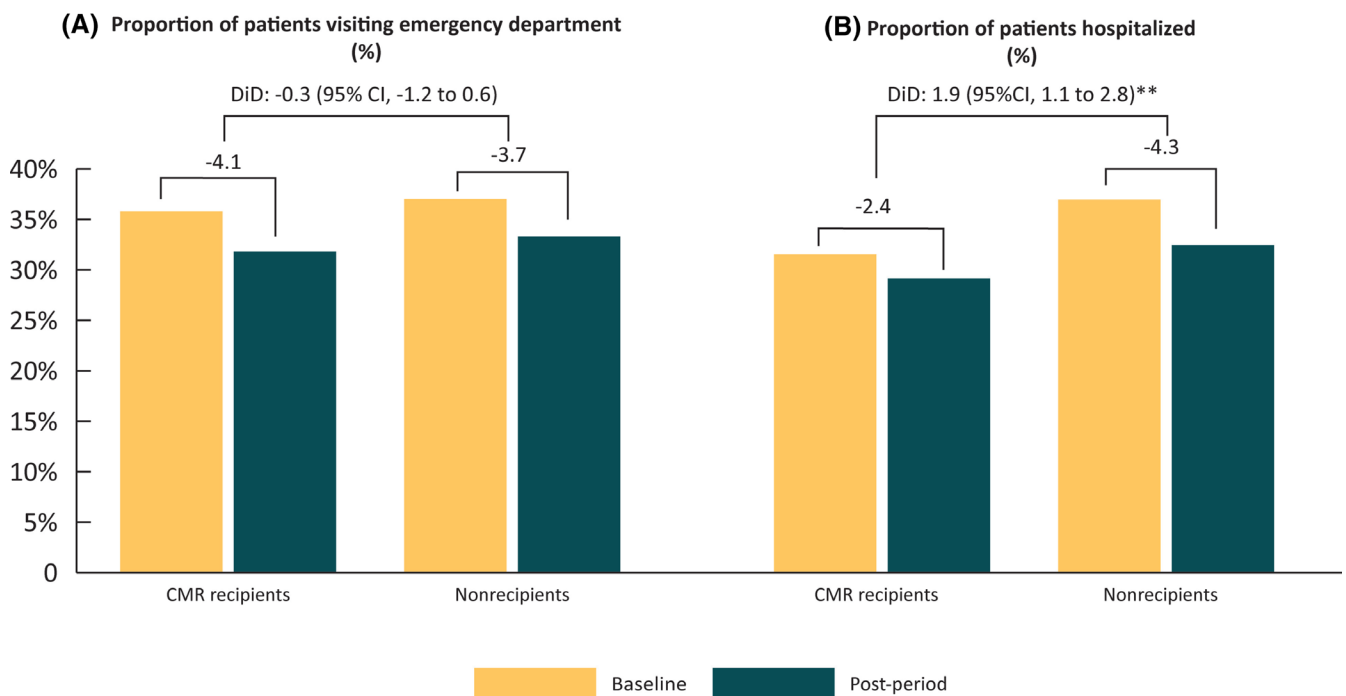


FIGURE 3 | Impact of Comprehensive Medication Review (CMR) on healthcare utilization. The figure presents the impact of CMR on two outcomes: (A) proportion of patients visiting emergency departments, and (B) proportion of patients hospitalized during the 365-day follow-up period. The difference-in-differences (DiD) estimates and confidence intervals (CIs) indicate the changes between CMR recipients and nonrecipients over time. ** p -value < 0.001.

in ED visits following CMR delivery ($\text{DiD} = -9.2$ visits per 1000 patients [95% CI, -28.7 to 10.4]).

3.4 | Hospitalization

The proportion of patients who experienced a hospitalization during the 1-year baseline period was 31.5% for CMR

recipients and 36.7% for nonrecipients. This proportion decreased by 2.4% (95% CI, 1.7%–3.1%) among CMR recipients and by 4.3% (95% CI, 3.8%–4.8%) among nonrecipients during the post-period. The DiD estimate suggests a 1.9% (95% CI, 1.1%–2.8%) higher probability of hospitalization among CMR recipients relative to nonrecipients (Figure 3). The DiD estimate for the count of hospitalizations suggests 42.0 (95% CI, 22.5–61.6) more hospitalizations for every 1000

TABLE 2 | Subgroup analysis of oral antidiabetic drugs adherence.

Characteristics	Subgroups	DiD estimates (95% confidence interval)	
		Proportion of days covered (%)	Proportion of adherent (%)
Age	≤ 75	2.4 (1.8 to 3.0)	1.4 (0.1 to 2.8)
	> 75	1.8 (1.1 to 2.6)	2.3 (1.3 to 3.3)
Sex	Male	2.9 (2.2 to 3.5)	1.9 (0.6 to 3.1)
	Female	2.3 (1.8 to 2.9)	2.1 (1.0 to 3.1)
Race/ethnicity	Non-Hispanic White	3.0 (1.6 to 4.5)	1.7 (0.7 to 2.6)
	Black/African American	1.3 (−1.7 to 4.3)	2.4 (0.1 to 4.8)
	Asian/Pacific Islander	2.8 (1.3 to 4.4)	2.3 (−2.5 to 7.0)
	Hispanic	1.8 (−5.5 to 9.1)	3.5 (1.0 to 6.0)
	American Indian/Alaska Native	2.6 (−1.6 to 6.7)	−3.4 (−16.3 to 9.4)
Geographic region	Northeast	2.1 (0.9 to 3.3)	0.9 (−0.9 to 2.8)
	South	2.8 (2.1 to 3.6)	2.7 (1.4 to 3.9)
	Midwest	2.7 (1.8 to 3.7)	2.2 (0.7 to 3.8)
	West	1.1 (−0.2 to 2.4)	0.9 (−1.2 to 3.0)
	Other	0.3 (−10.6 to 11.3)	5.5 (−13.3 to 24.3)
Urban residence	Yes	2.2 (1.7 to 2.8)	1.7 (0.8 to 2.6)
	No	3.2 (2.2 to 4.2)	3.0 (1.3 to 4.7)
Income status	Medicaid enrollment	3.1 (2.1 to 4.0)	2.2 (0.7 to 3.7)
	QMB enrollment	2.7 (0.4 to 5.0)	4.0 (−0.3 to 8.1)
	SLMB/QI enrollment	1.5 (−1.6 to 4.6)	0.1 (−5.0 to 5.3)
	QDWI individual	3.0 (−0.8 to 6.9)	3.4 (−9.6 to 9.8)
	Non-Medicaid or unknown	2.1 (1.5 to 2.7)	1.8 (0.8 to 2.8)

Abbreviations: DiD, difference-in-differences; QDWI, qualified disabled and working individuals; QI, qualifying individual; QMB, qualified Medicare beneficiary; SLMB, specified low-income Medicare beneficiary.

CMR recipients relative to nonrecipients after CMR delivery (Data S1).

3.5 | Subgroup and Sensitivity Analyses

Subgroup analyses revealed that the impact of CMR was consistent across various demographics, including age, sex, geographic region, urban residency, and income status, showing no significant differences in these subgroups. However, certain subgroups did not experience the same significant benefits. For instance, among patients identified as Black or African, we did not observe a significant improvement in PDC (DiD: 1.3% [95% CI, −1.7% to 4.3%]) (Table 2). Black or African American patients similarly did not experience significant improvements in the proportion of statin utilization attributable to CMR exposure (DiD: 1.5% [95% CI, −0.6% to 3.7%]) (Data S1).

When adjusting the threshold for defining treatment adherence to a PDC of 0.9, the results continued to demonstrate a positive effect of CMR on adherence (DiD: 1.1% [95% CI, 0.3%–2.0%]). However, CMR did not provide significant benefits regarding

TABLE 3 | Results of sensitivity analyses.

Sensitivity analyses	DiD (95% confidence interval)
Proportion of adherent (PDC ≥ 0.9) (%)	1.1 (0.3–2.0)
Proportion of patients visiting emergency department within 180 days (%)	1.6 (0.9–2.4)
Number of emergency department visits within 180 days (%)	2.4 (1.1–3.8)
Proportion of patients hospitalized within 180 days (%)	4.2 (3.5–5.0)
Number of hospitalizations within 180 days (%)	7.5 (6.3–8.8)

Abbreviations: DiD, difference-in-differences; PDC, proportion days covered.

the 180-day ED visit rate or frequency, nor the 180-day hospitalization rate or frequency. The details of these sensitivity analyses can be found in Table 3.

4 | Discussion

In what we believe to be the largest evaluation of the potential benefits of MTM as part of the Part D benefit in patients with T2DM, this retrospective cohort study suggests significant improvements from CMR delivery on medication quality measures, including improved medication adherence of OAD and statin utilization. We further found an increase in rates of hospitalization 1 year after CMR delivery.

The results of this study add to a growing body of evidence of the benefits of engaging pharmacists to manage patients with chronic health conditions. One study, utilizing electronic health records from a localized health system, demonstrated that the 121 patients with T2DM who engaged in pharmacist-led MTM were more likely to meet their HbA1c targets than those who did not receive such services [28]. An additional study examining the effectiveness of CMR among 4500 patients with heart failure, chronic obstructive pulmonary disease, and diabetes further demonstrated the benefits of MTM at improving diabetes adherence, despite variability in these benefits across organizations [16]. These studies suggest potential variability in the benefits of CMR based on how the service is delivered and the populations who receive MTM. These will be important considerations for future studies, particularly in light of upcoming changes to the MTM program. These changes will relax MTM eligibility for Contract Year 2025 by including HIV/AIDS as a core disease and considering all Part D maintenance drugs in the MTM eligibility criteria, which is expected to significantly increase the number of patients eligible for MTM services [29].

To further investigate whether CMR interventions impact healthcare utilization, we compared the proportion of ED visits and hospitalizations between CMR recipients and nonrecipients. We found that CMR interventions did not lead to a reduction in ED visits or hospitalizations 1 year after CMR delivery. Contrary to our initial hypothesis, hospital visits increased following a CMR encounter. Since the current MTM program only requires one annual CMR, its benefits may not be fully realized after a single visit, suggesting that CMS should consider more frequent CMR delivery. Additionally, improved medication adherence and the resulting clinical outcomes such as A1C control may take longer than 1 year to produce significant clinical benefits, highlighting the need for long-term evaluations [30]. The modest impact of CMR on OAD adherence could also be influenced by pharmacist recommendations to discontinue inappropriate medications or adjust treatments due to side effects, which may lower adherence rates.

The impact of the recent, and rapid, adoption of GLP-1 receptor antagonists is an area for future consideration. Given shifts in patient and prescriber preferences for these products, this has the potential to influence the decision of patients to switch from oral to injectable treatment options during CMR encounters. The period of observation in our study precedes the majority of this uptake and is not likely to have a significant influence on our study results [31]. Future studies should account for this factor when assessing changes in antidiabetic medication adherence.

4.1 | Clinical Implication

Our study leveraged the Medicare MTM database to demonstrate that T2DM patients who received a CMR service had significantly higher OAD adherence compared to those who did not receive a CMR. Maintaining adherence is challenging for many patients and must be tailored to each individual based on their own personal preference.

Pharmacists, as key healthcare providers, play a vital role in this process [32]. One of the central tenets of the Pharmacist Patient Care Process, which has been endorsed by all major pharmacy organizations as the standard of practice for pharmacists providing patient care, is the consideration of a patient's overall health goals and priorities during medication assessment and planning [32]. Pharmacists can play a key role in helping patients improve medication adherence during CMR encounters by using a patient-centered process and through shared decision making and collaboration with a patient's entire health care team [32]. They simplify medication regimens, reduce pill burden, and provide personalized guidance on medication timing to help patients integrate treatments into their daily routines [33]. Additionally, they enhance adherence by understanding and addressing patients' social determinants of health, providing tailored counseling and support, and fostering a strong patient-pharmacist relationship [34]. Our study demonstrates that their involvement significantly improves treatment adherence, which is considered one of the most common problems in diabetes care [35].

Despite the potential benefits derived from a CMR encounter, as of 2020, it is estimated that only 35% of eligible patients actually received an annual CMR through the Part D program [36]. The onus to deliver CMRs currently falls to the Part D plans. In 2016, the completion of an annual CMR was incorporated as a performance incentive in the Medicare Star Ratings program. However, the low rate of CMR completion demonstrates an opportunity for providers to refer patients to pharmacists for these services, given the evidence of benefit. Therefore, based on our findings, we strongly recommend integrating MTM pharmacists into the diabetes care team to optimize patient outcomes.

4.2 | Strengths and Limitations

This study represents, to the best of our knowledge, the first observational study to use a large comprehensive national sample of Medicare Part D claims data to assess the impact of CMR on OAD adherence and statin utilization among beneficiaries with T2DM. Consequently, our findings are more generalizable to older adults with T2DM than prior studies. However, this study is not without limitations. First, the PDC is an indirect measure of adherence derived from pharmacy fill histories and may overestimate true medication adherence due to its inability to directly observe actual medication-taking behavior. Second, while approaches like IPTW effectively balance observed covariates across groups, they cannot account for unmeasured confounders, which could influence our interpretation of CMR's effectiveness. Notably, our study is unable to control for baseline or follow-up rates of diabetes control as it lacks laboratory data

such as A1C information which would be required to make such inferences. Third, by including only participants aged over 66, our study's findings may not be applicable to younger individuals with T2DM, a notable consideration since the disease also affects these populations. Future research should aim to address this limitation. Fourth, we did not study the cumulative effect of CMR. Future studies should focus on whether multiple CMR exposures have larger impact on these outcomes. Fifth, our study did not assess the financial impact of CMR. Future research should explore whether CMR services contribute to cost savings in healthcare utilization. Lastly, our study excluded injectable antidiabetic medications, such as insulin and GLP-1 receptor agonists, which may limit the generalizability of our findings to the broader T2DM population. Given the increasing use of GLP-1 receptor agonists following their expanded indications for obesity and other conditions, future research should explore the impact of CMR services on adherence and outcomes among patients using these therapies [37].

5 | Conclusions

Despite the mandate to cover MTM services as part of the Medicare Part D benefit beginning in 2006, to our knowledge, no comprehensive assessment of the effectiveness of the program in patients with diabetes has been conducted to date. The results of our study suggest significant benefits to the quality of medication use as measured by medication adherence and the initiation of statin treatments among patients who received a CMR. The results point to benefits of engaging pharmacists as members of the health care team and additional opportunities to study the impact of pharmacy services on the health of patients with diabetes.

Author Contributions

Tsung-Hua Shen contributed to the study conception, design, data analysis, results interpretation, manuscript drafting, editing, reviewing, and submission. Felix Cheuk Wun Ting contributed to the study design, data analysis, coding quality check, results interpretation, manuscript editing, and approval. Shaquib Al Hasan contributed to the study design, results interpretation, manuscript reviewing, editing, and approval. Joel F. Farley contributed to the study conception, design, results interpretation, manuscript reviewing, editing, and approval.

Conflicts of Interest

Joel F. Farley has received consulting support from AstaLynx Global. The remaining authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.